

Case Study - Cross-Asset Microstructure Alpha with a Domain-Shift Fallback

A recruiter-facing summary of the Phase 4 project: real LOB evaluation, net bps after costs, and conservative behavior under transfer risk.

0.1711	29195	9/10	3.70
CAMP-R mean net bps / step	Total net bps	Positive episodes	Episode t-stat

Why this project is useful

Most trading projects are hard to trust because they skip clean baselines, cost accounting, or robust out-of-sample protocols. This project was designed to be compact enough to audit and still strong enough to demonstrate real research quality.

Core design

- Microstructure-first signals rather than a black-box-only approach.
- Explicit spread-based costs and turnover-aware evaluation in net bps.
- Real AAPL walk-forward testing plus external Binance transfer.
- Hybrid method with a conservative fallback under domain shift.

Main comparison

CAMP-R = **0.1711**, RAMP-R = 0.1643, online_regime = 0.1431 mean net bps / step on the combined real benchmark.

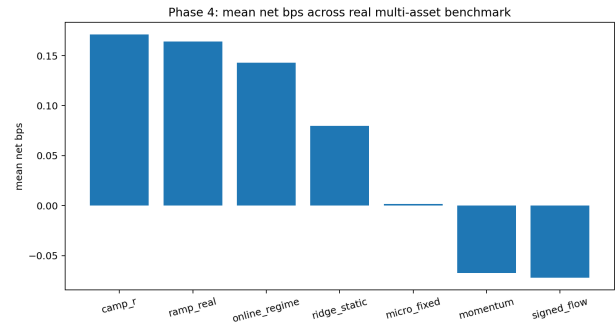
Asset split

AAPL / XNAS: **0.3148**
BNBUSDT / BINANCE: **0.0309**

Held-out honesty check

On the strict final AAPL block, online_regime leads at **0.6301**, while RAMP-R stays positive at **0.3971**. That makes the story more credible, not weaker.

Honesty note: this is a serious recruiting artifact built on real data and clean protocols, but it remains a compact benchmark. The correct positioning is research rigor + good experimental taste, not a claim of production-ready live alpha across a broad universe.



Best fit	Alpha researcher, quant researcher, execution-alpha hybrid
Novelty	Hybrid microstructure + ML + domain-shift fallback
What to claim	Research quality, benchmark discipline, robust communication - not production alpha at scale